2986 Words

1.

Consider a policy "A server authorises user's access if and only if the user is not blacklisted and the user either has a temporary pass or is an employee"

a.) Translate this policy into succinct first-order formula with predicates, connectives and quantifiers as least as possible

A - Authorise

B - Blacklist

T - Temp-pass

E - Employee

$\forall \boldsymbol{x}(\boldsymbol{A}(\boldsymbol{x}) \leftrightarrow \neg \boldsymbol{B}(\boldsymbol{x}) \wedge \big(\boldsymbol{T}(\boldsymbol{x}) \vee \boldsymbol{E}(\boldsymbol{x})\big))$

b.) Simplify the derived formula (in your answer to the previous question) into a conjunctive normal form, which is a conjunction ('AND') of disjunctive ('OR') clauses. Examples of such a form are: (☐ ∨ ☐ ∨ … ∨ ☐) ∧ (☐ ∨ ☐ ∨ … ∨ ☐) ∧ … ∧ (☐ ∨ ☐ ∨ … ∨ ☐

c.)

First Order Formula $\forall \boldsymbol{x}(\boldsymbol{A}(\boldsymbol{x}) \leftrightarrow \neg \boldsymbol{B}(\boldsymbol{x}) \wedge \big(\boldsymbol{T}(\boldsymbol{x}) \vee \boldsymbol{E}(\boldsymbol{x})\big))$
Let P =$(\boldsymbol{A}(\boldsymbol{x})$ Let Q = $(\neg \boldsymbol{B}(\boldsymbol{x}) \wedge \big(\boldsymbol{T}(\boldsymbol{x}) \vee \boldsymbol{E}(\boldsymbol{x})\big))$ 1. Transform all Biconditional statements into implications $P \leftrightarrow Q = (P \rightarrow Q) \wedge (Q \rightarrow P)$ $(P \rightarrow Q) \wedge (Q \rightarrow P) = (\neg P \vee Q) \wedge (\neg Q \vee P)$ $(\neg P \vee Q) \wedge (\neg Q \vee P)$ $\neg \boldsymbol{Q} = \neg (\neg \boldsymbol{B}(\boldsymbol{x}) \wedge (\boldsymbol{T}(\boldsymbol{x}) \vee \boldsymbol{E}(\boldsymbol{x})))$ $\neg \boldsymbol{P} = \neg(\boldsymbol{A}(\boldsymbol{x})$

2.

$$\boldsymbol{De\ Morgan's\ Law}$$
$$\neg(P \wedge Q) = \neg P \vee \neg Q$$
AND
$$\neg(P \vee Q) = \neg P \wedge \neg Q$$

We can simplify $\neg Q = \neg(\neg \boldsymbol{B}(\boldsymbol{x}) \wedge (\boldsymbol{T}(\boldsymbol{x}) \vee \boldsymbol{E}(\boldsymbol{x})))$ using the first rule shown above.

$\boldsymbol{B}(\boldsymbol{x}) \vee (\neg \boldsymbol{T}(\boldsymbol{x}) \wedge \neg \boldsymbol{E}(\boldsymbol{x}))$

$\neg P\ cannot\ be\ simplified\ further.$

3.

$$\boldsymbol{Central\ Law\ Theorem}$$
$$A \vee (B \wedge C) = (A \vee B) \wedge (A \vee C)$$

$(A(x)) \vee B(x) \vee (\neg T(x) \wedge \neg E(x))$

Using the Associative property, we can group elements together to fit the theorem, in this case both A & B represents A in the formula; B represents T and C represents E.

$(\neg \boldsymbol{Q} \vee \boldsymbol{P})$
A = $(A(x)) \vee B(x)$
B = $\neg T(x)$
C = $\neg E(x))$

$(A(x) \vee B(x) \vee \neg T(x)) \wedge (A(x) \vee B(x) \vee \neg E(x))$

So, we have $(\neg \boldsymbol{P} \vee \boldsymbol{Q}) \wedge (\boldsymbol{A}(\boldsymbol{x}) \vee \boldsymbol{B}(\boldsymbol{x}) \vee \neg \boldsymbol{T}(\boldsymbol{x})) \wedge \boldsymbol{A}(\boldsymbol{x}) \vee \neg \boldsymbol{E}(\boldsymbol{x})$

4.

$$\boldsymbol{Central\ Law\ Theorem}$$
$$A \vee (B \wedge C) = (A \vee B) \wedge (A \vee C)$$

$(\neg A(x) \vee (\neg B(x)) \wedge (T(x) \vee E(x))$

$(\neg \boldsymbol{P} \vee \boldsymbol{Q})$

A = ¬A(x)
B = ¬B((x)
C = T(x) & E(x)

$\boldsymbol{(\neg A(x) \vee \neg B(x)) \wedge (\neg A(x) \vee (T(x) \vee E(x))}$

4.

Everything is now in CNF, all that is to do is recombine parts.

$(\neg A(x) \vee \neg B(x)) \wedge (\neg A(x) \vee \big(T(x) \vee E(x)\big) \wedge\ (A(x) \vee B(x) \vee \neg T(x)) \wedge (A(x) \vee B(x) \vee \neg E(x))$

Finally place the quantifier back, the **answer** is as follows

$\forall \boldsymbol{x}\ \boldsymbol{(\neg A(x) \vee \neg B(x)) \wedge (\neg A(x) \vee \big(T(x) \vee E(x)\big) \wedge\ (A(x) \vee B(x) \vee \neg T(x)) \wedge (A(x) \vee B(x) \vee \neg E(x))}$

$\forall \boldsymbol{x}\ \boldsymbol{(\neg A(x) \vee \neg B(x)) \wedge (\neg A(x) \vee \big(T(x) \vee E(x)\big) \wedge\ (A(x) \vee B(x) \vee \neg T(x)) \wedge (A(x) \vee B(x) \vee \neg E(x))}$

2.

Consider a statement: 'No one has more than three grandmothers.' Translate this statement into a first-order formula using the propositional function $G(x, y)$, which represents 'x is the grandmother of y.' Justify your answer.

$\forall y\ \forall x1\ \forall x2\ \forall x3((G(x,y) \wedge G(x2,y) \wedge G(x3,y) \rightarrow (x1 = x2 \vee x1 = x3 \vee x2 = x3))$

3.

Let E be the set of even integers, O be the set of odd integers, and Z be the set of all integers. Determine each of these sets and express them in set notation. Justify your answer.

1. $\underline{E \cup O}$

   Roster Notation
   $E = \{\ldots -8, -6, -4, -2, 0, 2, 4, 6, 8 \ldots\}$
   $O = \{\ldots -7, -5, -3, -1, 1, 3, 5, 7 \ldots\}$
   $E \cup O = \{\ldots -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5 \ldots\}$

   Set Builder Notation
   $E = \{2n \mid n \in \mathbb{Z}\}$
   $O = \{2n + 1 \mid n \in \mathbb{Z}\}$

   $\mathrm{E} \cup \mathrm{O} = \{\mathrm{n} \mid n \in \mathbb{Z}\}$

   Justification – The union of E and O which is even and odd integers respectively will produce, a set containing all integers

2. $\underline{2. E \cap O}$

   Notation
   $E = \{\ldots -8, -6, -4, -2, 0, 2, 4, 6, 8 \ldots\}$
   $O = \{\ldots -7, -5, -3, -1, 1, 3, 5, 7 \ldots\}$

   $E = \{2n \mid n \in \mathbb{Z}\}$
   $O = \{2n + 1 \mid n \in \mathbb{Z}\}$

   $\mathrm{E} \cap \mathrm{O} = \{\}$
   or
   $\mathrm{E} \cap \mathrm{O} = \emptyset$

Justification – The intersection of E which consists of all even integers and O which consists of all odd integers, will leave nothing remaining, as even and odd numbers are opposites with no overlap, hence no intersection.

3. $\underline{Z - E}$

   Roster Notation
   $Z = \{\ldots -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5 \ldots\}$

$E = \{\ldots -8, -6, -4, -2, -0, 2, 4, 6, 8 \ldots\}$

$Z - E = \{\ldots -5, -3, -1, 1, 3, 5 \ldots\}$

Set Builder Notation

$Z = \{n \mid n \in \mathbb{Z}\}$
$E = \{2n \mid n \in \mathbb{Z}$

Z - E $= \{2n + 1 \mid n \in \mathbb{Z}\}$

Justification – Z is equal to all integers the subtraction of E which is equal to all even integers the process calculation of Z-E will leave only odd integers remaining as even ones will be removed from Z's set.

4.

Let h: $U \rightarrow \{0,1,2,3, \ldots, m-1\}$ be a function, where U is an arbitrary set and m is a positive integer. Can h be injective when $|U| > m$? Justify your answer

$Let\ h: U \rightarrow \{0,1,2,3 \dots, m-1\}$

$Where\ m\ is\ \mathbb{Z}^{+} < |U|$

$Let\ V\ =\ \{0,1,2,3, \dots\ , m-1\}$

$Let\ h : U \rightarrow V$

Assume function h is injective when |U| > m where U is an arbitrary set and m is a positive integer.

For all x in U there is at most one y in U such that h(y) = h(x)

Therefore, injectivity requires |U| ≤ |V| contradicting |U| > |V|

5.

Consider a binary tree G = (V, E) where both V and E are empty. Write a proof to show that G is a full tree.

$Let\ V\ =\ \emptyset$

$Let\ E\ =\ \emptyset$

$Let\ Binary\ Tree\ G\ = \{V, E\}$

Since there are no edges $(E\ =\ \emptyset)$ or nodes $(V\ =\ \emptyset)$, therefore every node has either 0 or 2 children and Binary Tree G is a full tree.

6.

Consider the function max(x, y) which returns either the real number x or the real number y whichever one is the largest, and function min(x, y) that returns the smallest value. Write a proof-by-cases to show that max(x, y) + min(x, y) = x + y

Let x, y : $\mathbb{R}$

Let max(x,y) = {x if ≥ y, y if y > x.}

Let min(x,y) = {x if ≤ y, y if y < x.}

Case 1: x > y

*Assuming x > y, max(x,y) = x ,**AND** min(x,y) = y. Therefore min(x,y) + max(x,y) = x + y*

Case 2: x ≤ y

*Assuming x ≤ y, max(x,y) = y ,**AND** min(x,y) = x. Therefore min(x,y) + max(x,y) = x + y*

Conclusion

$Therefore\ by\ cases\ when\ x, y \in \mathbb{R},\ max(x, y) + min(x, y) = x + y.$

7.

For a function which evaluates an arbitrary list of size n which compares a list of comparable objects will the function always return True if x is in the list otherwise False?

Base Case :

$Xs = [] \Rightarrow Contains(Xs, x) = False$

When Xs = [] the function will evaluate to False and exit the programme prematurely, working as intended as element x cannot be within an empty list.

```
If not Xs: #Xs is equivalent to [] here.
      Return False #If Evaluated to False this is an early return
If Xs[0] == x: #This line will not run
      return True
return contains(Xs[1:], x)
```

Induction Step :

Assuming that Xs at index 0 is equal to Y

$$\boldsymbol{Assume\ Xs[0] == y}$$

Case1:

$\boldsymbol{y} == x \Rightarrow Contains(Xs, x) = True$

```
If Xs[0] == x: #This line will evaluate to True
      return True #True will be returned
return contains(Xs[1:], x) #This line will be skipped.
```

Because Xs[0] == x the function will return True and exit the programme

Case2:

$\boldsymbol{y} ! = x \Rightarrow Contains(Xs, x) = Contains(Xs[1:], x)$

```
If Xs[0] == x:
      return True #This line will be skipped
return contains(Xs[1:], x) This line will be ran
```

Because Xs[0] != x the function will skip the return True and continue from the last line shown above, comparing every element within the list to x.

$$\boldsymbol{return\ contians(Xs[1:], x)}$$

8.

Consider the weighted directed graph below representing a data-centre network. Perform Dijkstra's algorithm finding a path from Reno to Sacramento. Show each step of the execution of the algorithm.

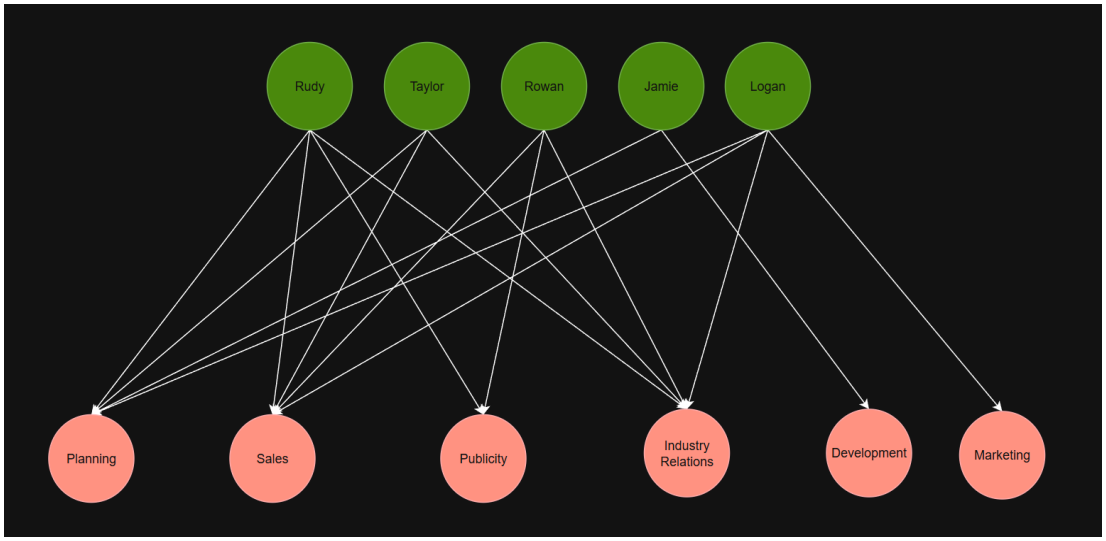
Explanation	Current Observed Values (units)	Current Best Estimate (units)
Initial node is explored with a value of 0, all other nodes are unexplored with a unit value of X (0) – Represents Unexplored (1) – Represents Explored	Reno = 0 (0)	**Reno**(0) = 0 Bakersfield(0) = X Las Vegas(0) = X Sacramento(0) = X San Francisco(0) = X Los Angeles(0) = X
From the initial node, all immediate paths (connected nodes) are explored. These paths will attain a new unexplored estimate. **Reno**	**Reno** -> **Bakersfield** = 133 (0+133) **Reno** -> **Las Vegas** = 331 (0+331) **Reno** -> **San Francisco** = 25 (0+25)	**Reno**(1) = 0 **Bakersfield**(0) = 133 **Las Vegas**(0) = 331 **San Francisco**(0) = 25 Sacramento(0) = X Los Angeles(0) = X
The next vertex explored, will be the lowest current unexplored estimate **(San Francisco)** The estimate for **Las Vegas** & **Los Angeles** is updated as a new shortest path is found.	**San Francisco** -> **Las Vegas** = **112** (0+112) **San Francisco** -> **Los Angeles** = **92** (0+92) **Reno** -> San Francisco -> **Las Vegas** = **137** (0+25+112) **Reno** -> San Francisco -> **Los Angeles** = **117** (0+25+92)	**Reno**(1) = 0 **San Francisco**(1) = 25 **Bakersfield**(0) = 133 **Los Angeles**(0) = 117 **Las Vegas**(0) = 137 Sacramento(0) = X
Next Smallest Unexplored Town **(Los Angeles)** New Estimate for Sacramento	**Los Angeles** -> **Sacramento** = 92 (0+92) **Los Angeles** -> **Bakersfield** = 59 (0+59) **Reno** -> San Francisco -> Los Angeles -> **Sacramento** = 209 (0+25+92+92) **Reno** -> San Francisco -> Los Angeles -> **Bakersfield** = 176 (0+25+92+59)	**Reno**(1) = 0 **San Francisco**(1) = 25 **Los Angeles**(1) = 117 **Bakersfield**(0) = 133 **Sacramento**(0) = 209 **Las Vegas**(0) = 137
Next Smallest Unexplored Town **(Bakersfield)** No values will be updated as there are no improved estimates.	**Bakersfield -> Reno** = (0+109) **Bakersfield -> San Francisco =** (0+127) **Reno** -> Bakersfield -> **Reno** = 242 (0+133+109) **Reno** -> Bakersfield -> **San Francisco** = 260 (0+133+127)	**Reno**(1) = 0 **San Francisco**(1) = 25 **Los Angeles**(1) = 117 **Bakersfield**(1) = 133 **Sacramento**(0) = 209 **Las Vegas**(0) = 137
Next Smallest Unexplored Town **(Las Vegas)**	**Las Vegas** -> **Sacramento** = 175 (0+175) **Las Vegas** -> **Reno** = 313 (0+313)	**Reno**(1) = 0 **San Francisco**(1) = 25

No values updated	**Reno** -> San Francisco -> Las Vegas -> **Sacramento** = 312 (0+25+112+175) **Reno** -> San Francisco -> Las Vegas -> **Reno** = 450 (0+25+112+313)	**Bakersfield**(1) = 133 **Las Vegas**(1) = 137 **Los Angeles**(1) = 92 **Sacramento**(0) = 209
Next Smallest Town **(Sacramento)** No values updated	**Sacramento** -> **San Francisco** = 92 (0+92) **Sacramento** -> **Los Angeles** = 114 (0+114) **Reno**-> San Francisco -> Los Angeles -> Sacramento -> **San Francisco** = 301 (0+25+92+92+92) **Reno**-> San Francisco -> Los Angeles -> Sacramento -> **Los Angeles** = 323 (0+25+92+92+144)	**Reno**(1) = 0 **San Francisco**(1) = 25 **Bakersfield**(1) = 133 **Las Vegas**(1) = 137 **Los Angeles**(1) = 117 **Sacramento**(1) = 209
Finally, all paths are explored, the lowest cost path to Sacramento is the **Reno** —> San Francisco —> Los Angeles -> **Sacramento (209)**		

9.

A.) Draw a graph to model all possible assignments that are suitable for the skills of the members of staff. Use the drawing to justify whether the graph is bipartite.

Figure 1A

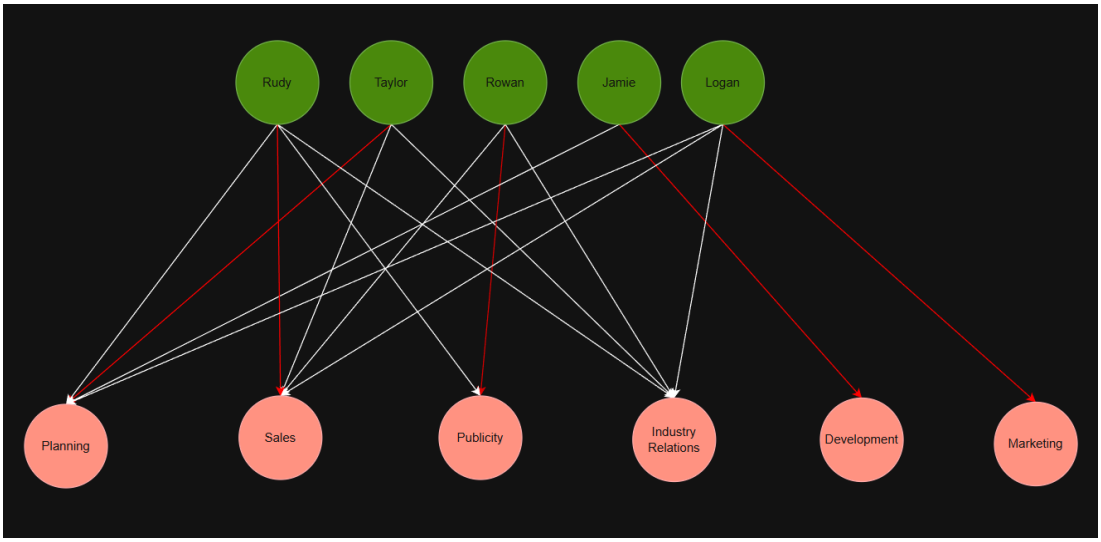


(Draw.io, n.d)

The Graph in **Figure 1A**, is bipartite because its nodes can be divided into two disjoint sets, such that every edge connects a node in one set to the other, with no edges between elements of the same set.

B.) Draw a bipartite matching such that each staff member is assigned a task that they are skilled in. Use the drawing to justify whether every task is assigned a staff.

Figure 1B



(Draw.io, n.d)

The Graph in **Figure 1B,** demonstrates a bipartite matching (shown in red); the matching assigns every staff to a task they're skilled in. Although not all tasks can be assigned as the matching saturates all staff nodes but not all task nodes.

Therefore, all tasks cannot be assigned a corresponding staff.

10.

Take the derivatives of the following with respect to x. Show your step-by-step calculation.

1. $\mathbf{3x^5 - 2x + 7}$

To reach the derivative, the power rule must be used, the power rule states that the coefficient is to be multiplied against the indices and, the indices multiplied is to be decreased by 1.

	Rule (Power Rule)	Question
Function	x^n	$3x^5 - 2x + 7$
Derivative	nx^{n-1}	$2x^4 - 2$

Question $\mathbf{3x^5 - 2x + 7}$
$\mathbf{1}.\mid 3x^5 \rightarrow 15x^4$
$\mathbf{2}.\mid \ -2x \rightarrow -2$
$\mathbf{3}.\mid \ +7 \rightarrow 0$
$\mathbf{4}.\mid \mathbf{15x^4 - 2}\ (\boldsymbol{Recombined\ parts})$

<u>Answer is $\mathbf{15x^4 - 2}$</u>

2. $\frac{\mathbf{3\sqrt{x}+2}}{\mathbf{x^2+1}}$

To reach the derivative of this, the quotient rule must be used, which states the numerator is to be f and the denominator g. The derivative of f multiplied by g subtracted by the derivative of g multiplied by f all divided by g squared.

	Rule (Quotient Rule)	Question
Function	f/g	$\boldsymbol{\frac{3\sqrt{x}+2}{x^2+1}}$
Derivative	$\frac{f'g - g'f}{g^2}$	$\frac{\frac{3(x^2+1)}{2\sqrt{x}} - 2x(3\sqrt{x}+2)}{(x^2+1)^2}$

Question
$\boldsymbol{\frac{3\sqrt{x}+2}{x^2+1}}$

1. | $f = 3\sqrt{x} + 2$ $g = x^2 + 1$ $g^2 = (x^2 + 1)^2$ $g' = 2x$
2. $f = 3\sqrt{x} + 2 = 3x^{\frac{1}{2}}+2$ $f' = 3\frac{1}{2}x^{-1/2} = \frac{3}{2\sqrt{x}}$
$$\frac{f'g - g'f}{g^2} = \frac{\frac{3}{2\sqrt{x}}x^2 + 1 - 2x\,3\sqrt{x} + 2}{(x^2+1)^2}$$ **<u>Answer =</u>** $$\frac{\frac{3(x^2+1)}{2\sqrt{x}} - 2x(3\sqrt{x}+2)}{(x^2+1)^2}$$ The answer can be simplified further, although the <u>summative does not require it.</u>

11.

Consider the throughput of a distributed concurrent system is modelled by an equation $T(x) = -x^6 + 3x^4 - 2x^2 + 1$ *where* $x \geq 0$ *is the level of concurrency*

1.) $-x^6 + 3x^4 - 2x^2 + 1$

2.) $-6x^5 + 12x^3 - 4x$

3.) $-2x(3x^4 - 6x^2 + 2)$

4.) $3x^4 - 6x^2 + 2 = 0$

5.) $3(t^2 - 6t + 2) = 0$

6.)

Quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

A = 1
B = -6
C = 2

$$\frac{-(-6) \pm \sqrt{(-6)^2 - 4(3)(2)}}{-3(2)}$$

T Values
$Positive = -1.577350269$
$Negative = -0.4226497308$

7.) $X^2 = T$
$X = \sqrt{T}$

We will have 4x values

$x = -\sqrt{-1.577350269} = -1.25592606$

$x = \sqrt{-1.577350269} = 1.25592606$

$x = -\sqrt{-0.4226497308} = -0.6501151673$
$x = \sqrt{-0.4226497308} = 0.6501151673$

8.)

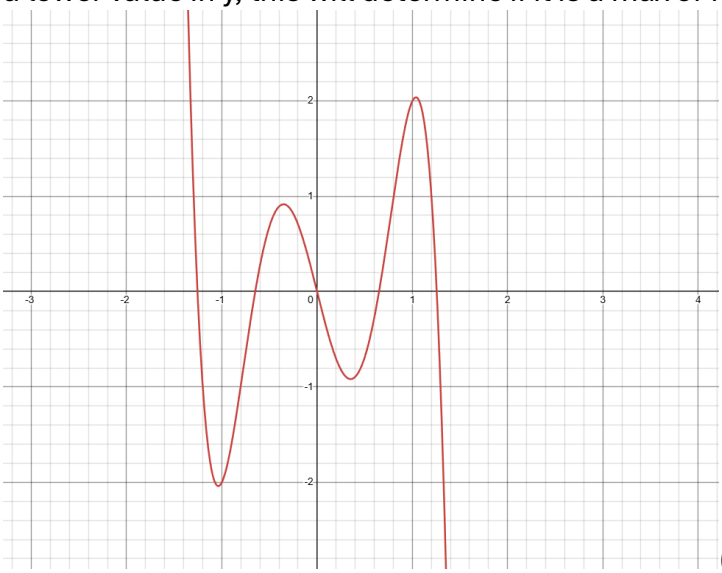
$-6(x + 0.6501151673)(-0.6501151673)(x + 1.25592606)(x - 1.25592606)$

Reverse signs

$+0.6501151673$
-1.25592606
$+1.25592606$

9.)

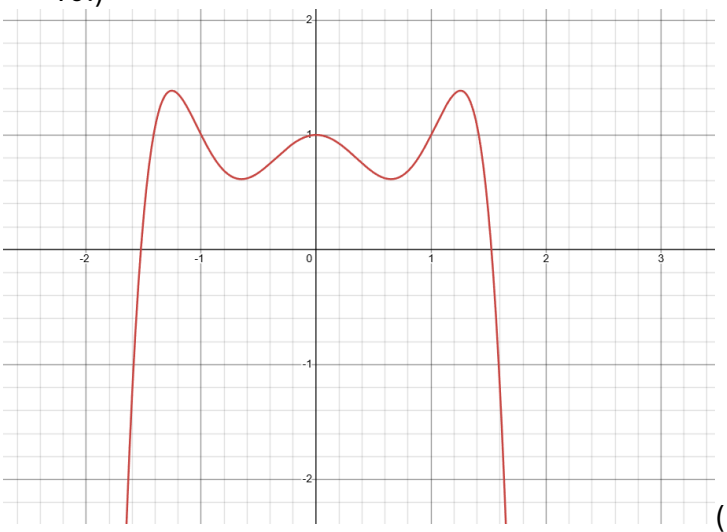
Observe the graph for change in derivative where it goes from a higher value in y to a lower value in y, this will determine if it is a max or min of a function.



(Desmos, n.d)

The first negative to positive occurs at $x = -1.25592606$

10.)



(Desmos, n.d)

$T(-1.25592606)$
$= -(1.25592606)^6 + 3(-1.25592606)^4 - 2(-1.25592606)^2 + 1$

$\underline{\boldsymbol{Final\ Answer}\ =\ 1.384900179}$

12.

Suppose you have measured API responses times (milliseconds) in one hour:

a. Calculate mean

To calculate mean, we add the sum of all numbers divided by the number of total values in the set.

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} xi \qquad \bar{x} = \frac{\sum x}{n}$$

1. $\frac{75.43 + 133.36 + 122.49 + 83 + 90.29 + 100.34 + 130.47 + 124.8 + 140.92 + 90.4 + 52.83 + 141.22 + 117.8 + 133.18 + 107.95 + 89.24 + 131.49 + 89.04 + 120.77 + 124.6 + 117.06 + 137.72 + 82.18 + 96.41 + 166.19 + 98.73 + 98.62 + 86.48 + 78.1 + 103.37 + 85.09 + 90.29 + 66.51 + 109.18 + 105.45}{35}$

2. $\frac{3721}{35}$

3. $\frac{3721}{35} = 106.3142857142857142857142857142 9$

4. 106.31 to 2.dp

b. Calculate median

To find the median, the data is arranged in ascending order, if the number of values is odd the median is the middle value, if it is even, the median is the mean of the two central values.

52.83
66.51
75.43
78.1
82.18
83
85.09
86.48
89.04

89.24
90.29
90.29
90.4
96.41
98.62
98.73
100.34
$\underline{103.37}$ - Median
105.45
107.95
109.18
117.06
117.8
120.77
122.49
124.6
124.8
130.47
131.49
133.18
133.36
137.72
140.92
141.22
166.19

The median is **$\underline{103.37}$**

c. Calculate Standard Deviation

For this calculation I have rounded the mean from earlier to 106.31

Sample – Variance = **613.0685265**

$S^2 = \frac{\sum(xi-\bar{x})^2}{n-1}$ Sample Variance

$S = \sqrt{\frac{\sum(xi-\bar{x})^2}{n-1}}$ Sample Deviation

$((52.83 - 106.31)^2 + (66.51 - 106.31)^2 + (75.43 - 106.31)^2 + (78.1 - 106.31)^2 + (82.18 - 106.31)^2 + (83 - 106.31)^2 + (85.09 - 106.31)^2 + (86.48 - 106.31)^2 + (89.04 - 106.31)^2 + (89.24 - 106.31)^2 + (90.29 - 106.31)^2 + (90.29 - 106.31)^2 + (90.4 - 106.31)^2 + (96.41 - 106.31)^2 + (98.62 - 106.31)^2 + (98.73 - 106.31)^2 + (100.34 - 106.31)^2 + (103.37 - 106.31)^2 + (105.45 - 106.31)^2 + (107.95 - 106.31)^2 + (109.18 - 106.31)^2 + (117.06 - 106.31)^2 + (117.8 - 106.31)^2 + (120.77 - 106.31)^2 + (122.49 - 106.31)^2 + (124.6 - 106.31)^2 + (124.8 - 106.31)^2 + (130.47 - 106.31)^2 + (131.49 - 106.31)^2 + (133.18 - 106.31)^2 + (133.36 - 106.31)^2 + (137.72 - 106.31)^2 + (140.92 - 106.31)^2 + (141.22 - 106.31)^2 + (166.19 - 106.31)^2)$ / 34 = **613.0685265**

Sample – Deviation = **24.76022065**.

$$\sqrt{\mathbf{613.0685265}} = \mathbf{24.76022065}.$$

d. Calculate 95% Confidence Interval

$$Mean = 106.31$$
$$Standard\ Deviation = 24.760$$

$$\frac{a}{2} = \frac{1 - Confidence\ Level}{2} = \frac{1 - 0.95}{2} = \frac{0.05}{2} = 0.025$$

$$v = n - 1 = 35 - 1 = 34$$

$$t_{34,0.025} = 2.0322$$

$$\mu \rightarrow \bar{x} \pm t_{v,\frac{a}{2}} \frac{s}{\sqrt{n}}$$

$$\mu \rightarrow \overline{106.31} \pm 2.0322 \frac{24.760}{\sqrt{35}} = (114.8151713),\ (97.8048287)$$

95% Confidence interval is for the mean $(114.8151713),\ (97.8048287)$

e. **Compare and discuss the mean and median and which is a more appropriate representation of central tendency on this dataset? Justify your answer**

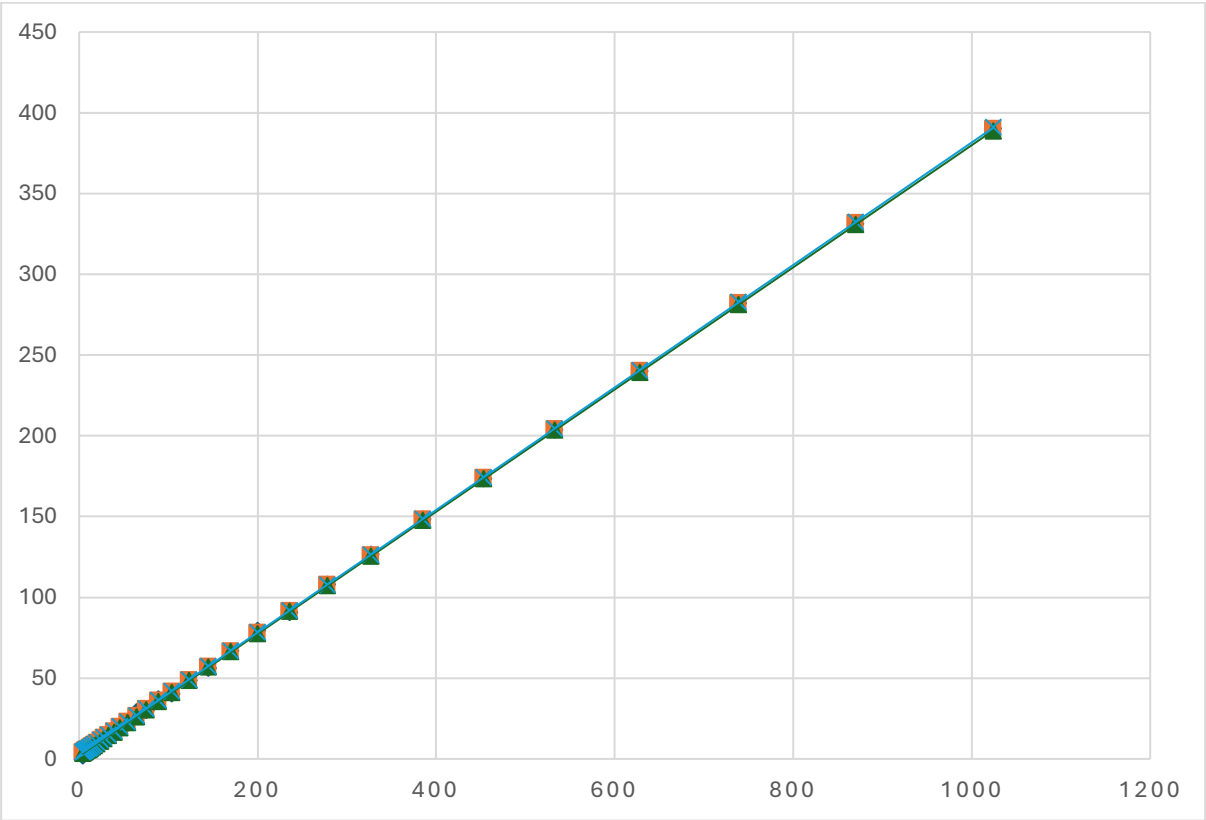
Mean is more useful, for symmetric standard distributions, as the mean is heavily penalized by outliers and skewed data. Median is more valuable when data is skewed as it will always be the central value. Either works for this dataset as the data is quite symmetric with low variance, the mean and median are virtually the same. Although I would lean towards mean due to the data being symmetric, and mean being quicker to calculate without computer aid as data does not have to be ordered for the mean.

13.

Suppose a series of measurement is made on the payload size (kilobyte) and I/O processing time (millisecond) during one hour of a server ingesting/egesting network traffic:

a. **Perform a least-square fitting on y = ax + b, where x is the payload and y is the I/O time and plot with 95% confidence interval on the fitted y.**

Figure 3A



(Microsoft, n.d)

The above graph demonstrates the 95% confidence interval, the green trendline represents the lower confidence interval and the blue the upper, the area in between is the 95% confidence interval. The orange dots represent predicted Y Values, and green dots represent actual Y values.

Not enough words remaining to show the full calculation, the formula and values are below, in chronological order; substitute for them. This question did not require step-by-step calculations.

$n = 35$
$\sum xy = 1440889.560$
$\sum y = 2642.38$
$\sum x = 6782$
$\sum x^2 = 3767402$

$$m = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2} = 0.378629989$$

$$b = \frac{\sum y - m\sum x}{n} = 2.128897645$$

$$\sum Residuals^2 = 30.0384177$$

$$\sqrt{\frac{\sum(Residuals)^2}{n-2}} = 0.954072891$$

$$\bar{x} = \frac{\sum x}{n} = 193.7714$$

$$\frac{a}{2} = \frac{1 - Confidence\ Level}{2} = \frac{1 - 0.95}{2} = 0.025$$

$t_{33,0.025=2.0345}$

$$Standard\ Error = \frac{s}{\sqrt{n}} = 0.161267753$$

$$SSxx = \sum x^2 - \frac{(\sum x)^2}{35} = 2453244.171$$

$$Confidence\ Interval\ = x = Y \pm t_a SE \sqrt{\frac{1}{n} + \frac{(x - x_m)^2}{SS_{xx}}}$$

Predicted Y	Lower Confidence	Upper Confidence
389.846	388.7661	390.9259
11.216	10.82625	11.60578
26.361	25.99582	26.72661
239.909	239.2783	240.5388
147.901	147.4967	148.3062
35.827	35.47411	36.17982
7.808	7.412453	8.204242
9.701	9.309037	10.09396
8.566	8.171095	8.96012
14.624	14.23983	15.00754
22.575	22.2039	22.94593
173.648	173.1891	174.1075
77.855	77.52671	78.18309
125.941	125.5736	126.3082
7.051	6.653801	7.448374
30.526	30.16654	30.88575
16.895	16.51542	17.27552
48.700	48.36077	49.04001
66.496	66.16658	66.82541
5.537	5.136465	5.936671

41.506	41.15997	41.85286
4.779	4.377781	5.180834
19.546	19.17013	19.92162
3.643	3.239736	4.047099
4.022	3.619087	4.425008
331.537	330.637	332.437
5.158	4.757124	5.558751
6.294	5.895138	6.692517
281.936	281.1853	282.6876
203.939	203.4054	204.472
57.030	56.69663	57.36387
91.486	91.15333	91.81782
12.731	12.34343	13.11765
4.401	3.998435	4.80292
107.388	107.0437	107.7323

b. Interpret a and b under the given context

A = The gradient which represents the change in time taken per unit of data, a high gradient demonstrates slow transfer and a shallow one demonstrates fast transfer.
B = The y-intercept which represents the overhead, or how much time is required before data transfer can occur.

c. Statistically evaluate the error / residuals

$Residual$	$Residual^2$
0.143994036	0.0207343
0.743982636	0.5535102
2.708783092	7.3375058
-0.138530479	0.0191907
0.468556751	0.2195454
1.223033377	1.4958106
0.791652533	0.6267137
-0.001497410	0.0000022
0.434392556	0.1886969
0.686312738	0.4710252
-0.444917022	0.1979512
-0.338282474	0.1144350
1.635104642	2.6735672

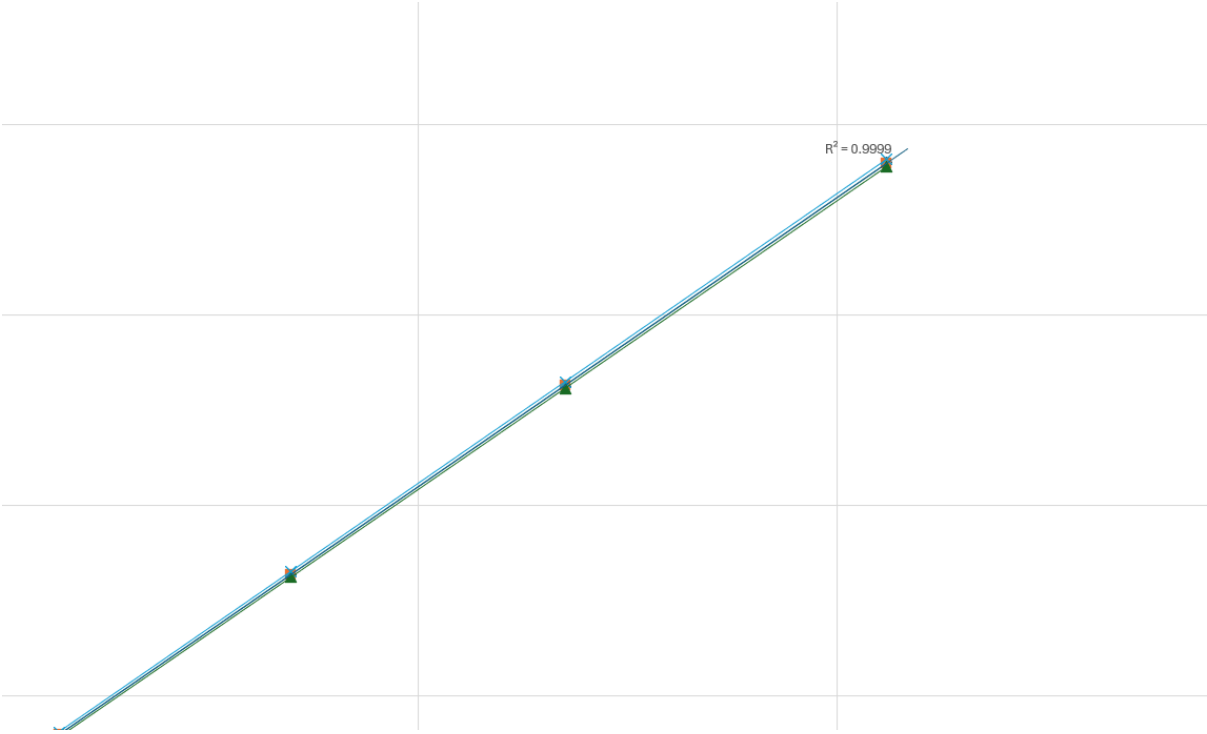
0.629096090	0.3957619
0.408912510	0.1672094
-0.446146783	0.1990470
-1.285467193	1.6524259
-0.250386236	0.0626933
0.234004300	0.0547580
0.853432465	0.7283470
-1.356416452	1.8398656
-0.809307558	0.6549787
-0.185877114	0.0345503
-1.763417592	3.1096416
0.207952419	0.0432442
-0.296987720	0.0882017
-1.957937547	3.8335194
-0.483827513	0.2340891
-0.156459213	0.0244795
-0.328681562	0.1080316
-1.170245985	1.3694757
-0.765574948	0.5861050
0.619462681	0.3837340
-0.290677570	0.0844934
0.681965531	0.4650770

$$\sqrt{\frac{\sum(Residuals)^2}{n-2}} = 0.954072891$$

$$Standard\ Error = \frac{s}{\sqrt{n}} = 0.161267753$$

The error rate is not very high; we can see this as the standard deviation for residuals is extremely narrow, demonstrates tighter predictions. Using graph **Figure 3A** we, can see this is supported as the predicted and actual values are practically overlapping.

Figure 4A



(Microsoft, n.d)

The above graph displays that the R^2 is 0.999 with the highest accuracy being a 1 and the lowest a 0, considering this; the predicted y values are extremely accurate.

Excel produced the R^2 value with the graph, I did not calculate it, so no R^2 formula will be provided

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